



04368258



EXAMINATIONS COUNCIL OF ZAMBIA



Examination for School Certificate Ordinary Level

Biology

5090/1

Paper 1 Multiple Choice

Tuesday

5 NOVEMBER 2024

Additional Materials:

- Multiple Choice Answer Sheet
- Soft clean eraser
- Soft pencil (type B or HB is recommended)

Time: 50 minutes

Marks: 40

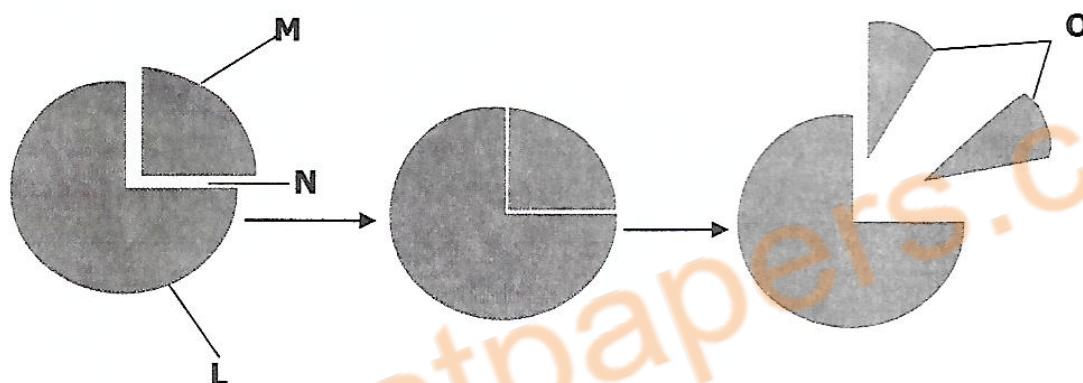
Instructions to Candidates

- 1 Ensure that the **school/centre name** and **subject paper, subject code, paper number, centre code, your examination number** and the **year** are correctly printed and shaded on the Answer Sheet.
- 2 There are **forty** questions in this paper. Answer **all** questions.
- 3 For each question there are four possible answers: **A, B, C** and **D**. Choose the **one** you consider **correct** and **shade** your choice using a **soft pencil** on the separate Answer Sheet.
- 4 **Read very carefully the instructions on the Answer Sheet.**

Information for Candidates

- 1 Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
- 2 Any rough working should be done in this Question Paper.
- 3 Cell phones and other electronic devices are **not allowed** in the examination room.

- 1 Which characteristic of living organisms involves ingestion, absorption and assimilation?
 - A Excretion
 - B Growth
 - C Nutrition
 - D Respiration
- 2 Which cell organelle is involved in respiration?
 - A Chloroplast
 - B Endoplasmic reticulum
 - C Mitochondrion
 - D Ribosome
- 3 Which of the following is true regarding active transport?
 - A An input of energy is not required for it to occur.
 - B It allows particles to pass easily between lipids in cell membranes.
 - C It occurs against a concentration gradient.
 - D Particles move from high concentration to low concentration.
- 4 The diagram demonstrates enzyme specificity and the stages in which an enzyme controlled chemical reaction may occur.



Identify L, M, N and O.

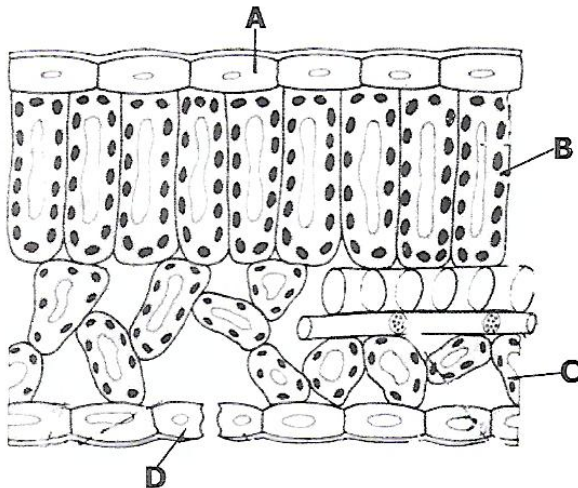
	L	M	N	O
A	Enzyme	Substrate	Active site	Products
B	Enzyme	Active site	Substrate	Products
C	Substrate	Enzyme	Active site	Products
D	Substrate	Active site	Enzyme	Products

- 5 Which of the following is **not** a good source of lipids?
 - A Avocado
 - B Castor seeds
 - C Caterpillars
 - D Cheese
- 6 Which of the following food nutrients promotes peristalsis?
 - A Carbohydrates
 - B Lipids
 - C Proteins
 - D Roughage

7 A diet rich in iron is ideal for which group of people?

- A Lactating women
- B Pregnant women
- C Pre-school children
- D The aged

8 The diagram shows part of a leaf.

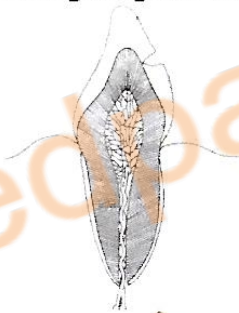


Which part shows epidermal cells?

9 Which of the following is an example of an organism which carries out saprophytic nutrition?

- A Amoeba
- B Mosquito
- C Rhizopus
- D Yeast

10 The following diagram shows a human tooth.



Identify the tooth and its function.

	Tooth	Function
A	Canine	for tearing flesh
B	Incisor	for biting off pieces of food
C	Molar	for grinding and crushing food
D	Premolar	for grinding and crushing food

11 Which of the following food nutrients is an end product of digestion along the alimentary canal?

- A Amino acids
- B Cellulose
- C Fats
- D Proteins

- 12** Which changes occur to the diaphragm and external intercostal muscles when a person breathes in deeply?

	Diaphragm	External intercostal muscles
A	Contracts	Contracts
B	Contracts	Relaxes
C	Relaxes	Contracts
D	Relaxes	Relaxes

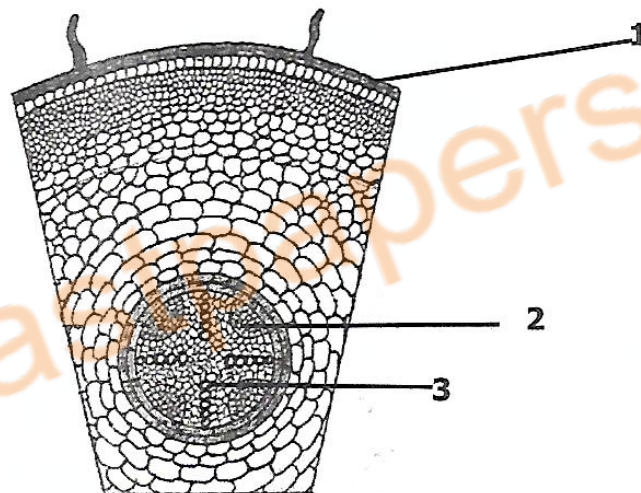
- 13** Which of the following is the correct word equation for anaerobic respiration in man?

- A** Glucose → carbon dioxide + water + energy.
- B** Glucose → ethanol + carbon dioxide + energy.
- C** Glucose → ethanol + energy.
- D** Glucose → lactic acid + energy.

- 14** Which of the following can help to reduce the causes of stigma to people living with Human Immunodeficiency Virus (HIV) and Acquired Immunodeficiency Syndrome (AIDS)?

- A** Counseling and testing
- B** Fear
- C** Myths
- D** Sensitisation

- 15** The following diagram shows the tissues in a section of the root of a green plant.



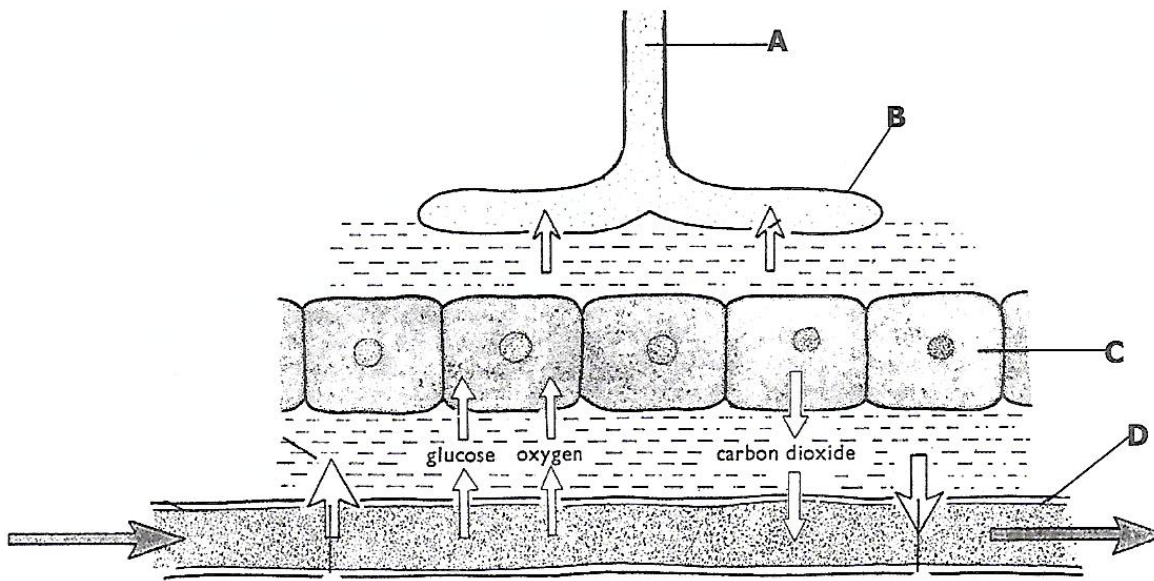
Identify the parts numbered 1, 2, and 3.

	1	2	3
A	Epidermis	Phloem	Xylem
B	Epidermis	Xylem	Phloem
C	Phloem	Epidermis	Xylem
D	Phloem	Xylem	Epidermis

- 16** Jane has blood group **A**. She needs a blood transfusion. From which blood groups can she receive blood?

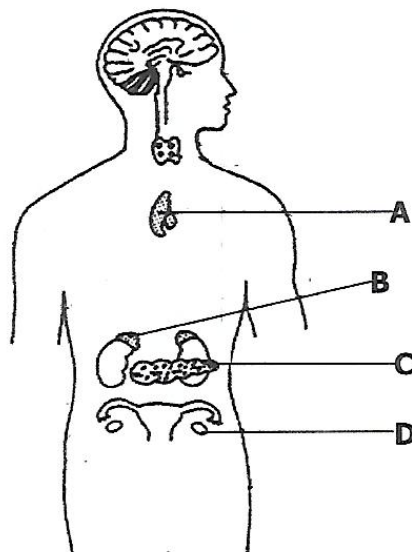
- A** A only
- B** AB only
- C** A and AB
- D** A and O

- 17** The following diagram shows the relationship between the blood circulatory system and the lymphatic system.

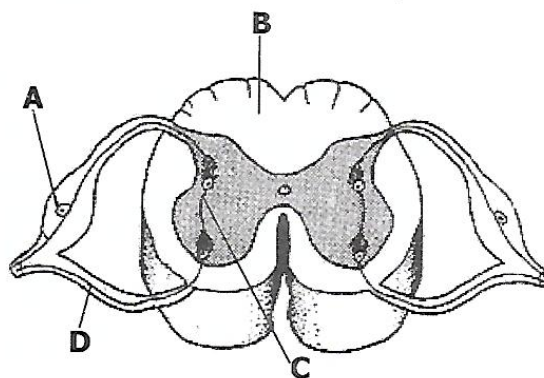


Which labelled part represents the lymph capillary?

- 18** Which of the following is an excretory product of the lungs?
- A Bile salts
 - B Carbon dioxide
 - C Oxygen
 - D Urea
- 19** Which of the following are the important organs in homeostasis?
- A Liver, heart and kidney
 - B Liver, kidney and skin
 - C Lungs, heart and sweat glands
 - D Lungs, skin and sweat glands
- 20** Which of the following labelled glands is associated to production of insulin?

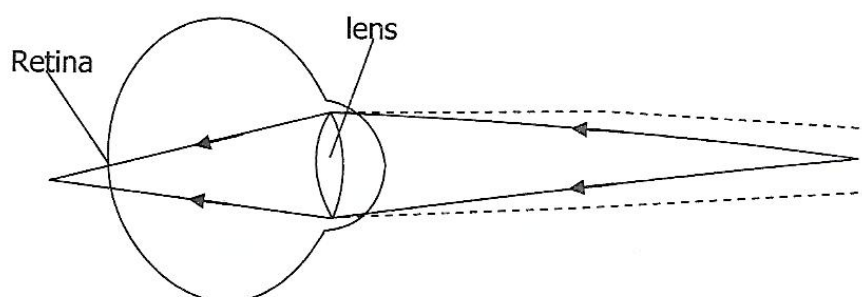


- 21 The diagram shows a cross section through a human spinal cord.



Which one of the parts labelled is a relay neurone?

- 22 The following diagram shows a defective eye of a person.



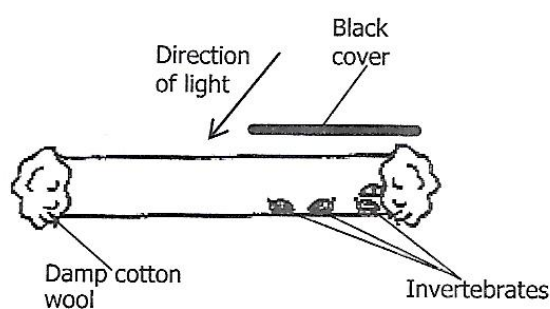
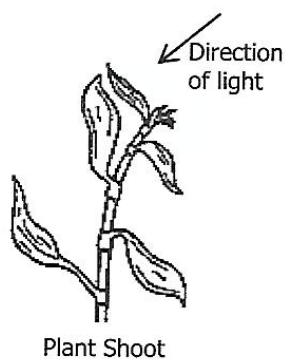
Identify the defect and how it could be corrected.

	Defect	Correction
A	Long sight	use of concave lens
B	Long sight	use of convex lens
C	Short sight	use of concave lens
D	Short sight	use of convex lens

- 23 The bones of the skull, ribs, sternum and vertebral column form up the ...

- A appendicular skeleton.
- B axil skeleton.
- C endo skeleton.
- D exo skeleton.

- 24 The diagram shows a plant shoot and invertebrates in relation to their response to a stimulus.



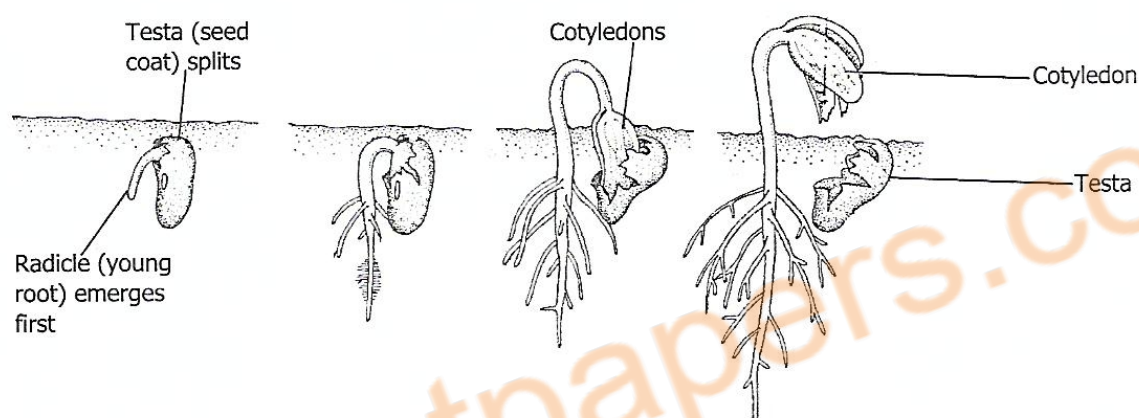
Which responses are being shown by these organisms?

	Plant Shoot	Invertebrates
A	Negative phototaxis	Positive phototropism
B	Negative phototaxis	Positive phototaxis
C	Positive phototaxis	Negative phototropism
D	Positive phototropism	Negative phototaxis

25 Which of the following differentiated cells is thickened with lignin and provide support to the stem of plants?

- A Collenchyma
- B Parenchyma
- C Phloem
- D Sclerenchyma

26 The diagram shows a type of germination.



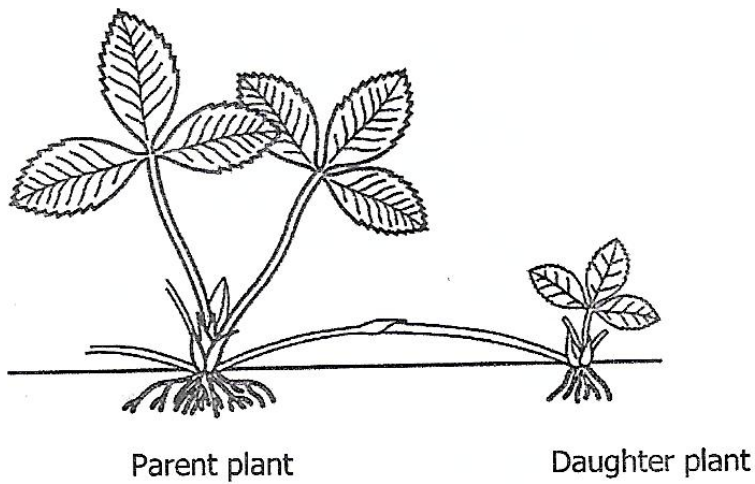
Which type of germination is shown in the diagram and why?

- A Epigeal because cotyledons are carried above ground.
- B Epigeal because testa remains in the ground.
- C Hypogeal because cotyledons are carried above ground.
- D Hypogeal because testa remains in the ground.

27 What type of asexual reproduction occurs in yeast?

- A Binary fission
- B Budding
- C Fragmentation
- D Spore formation

- 28 The following diagram represents one of the methods of natural propagation.



Which method of natural propagation is shown in the diagram?

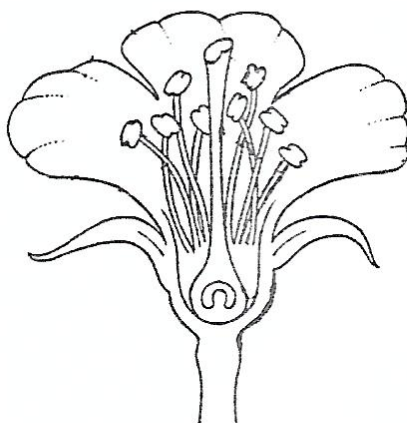
- A Corm
 - B Rhizome
 - C Runner
 - D Stem tuber
- 29 The diagram shows a generalised structure of a flower.



Identify the part labelled X.

- A Anther
- B Filament
- C Sepal
- D Stigma

- 30** The following diagram shows the structure of an insect pollinated flower.



Which feature on the diagram shows that the flower cannot self pollinate?

- A** Large anthers
- B** Presence of petals
- C** Presence of sepals
- D** Short filaments

- 31** Which of the following correctly states the functions of the prostate gland and seminal vesicles in the male reproductive system?

	Prostate gland	Seminal vesicle
A	Enables sperms to swim	Provides nutrients to sperms
B	Kills excess sperms	Enables sperms to swim
C	Kills excess ova	Kills excess sperms
D	Provides nutrients to sperms	Kills excess ova

- 32** Which of the following is a natural method of birth control?

- A** Condom
- B** Diaphragm
- C** Intra-uterine device
- D** Rhythm

- 33** The genetic characteristic expressed in an organism is called ...

- A** inheritance.
- B** mutation.
- C** trait.
- D** variation.

34 The following events take place in one of the stages of mitosis;

- The nuclear membrane disappears.
- The chromosomes move to the centre of the cell.
- The chromosomes align themselves at the equatorial region.
- The chromosomes are seen attached to the spindle fibres at the centromeres.

Which stage of mitosis is described in the statement?

- A** Anaphase
- B** Interphase
- C** Metaphase
- D** Prophase

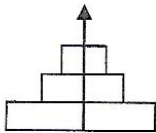
35 In classification, which of the following has the highest number of organisms?

- A** Class
- B** Genus
- C** Family
- D** Kingdom

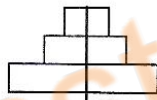
36 Which component of soil is lost when crop remains are burnt?

- A** Calcium
- B** Clay
- C** Humus
- D** Sand

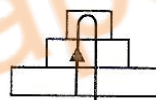
37 In which of the following pyramids of biomass does the arrow show the correct flow of energy?



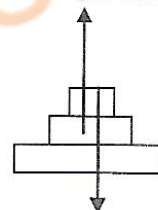
A



B



C



D

38 Which of the following factors directly causes population increase?

- A** Diseases
- B** Emigration
- C** Immigration
- D** Wars

39 Which of the following is likely to contribute to land pollution?

- A** Dead plant remains
- B** Mineral salts
- C** Non-biodegradable materials
- D** Soil microorganisms

- 40** Which of the following is the importance of biodiversity?
- A** Competition between carnivores and herbivores.
 - B** Depletion of wetland reserves.
 - C** Equilibrium of organisms in a given locality.
 - D** Variety of trees for charcoal production.

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